



A Comparative Analysis of Open Source Business Intelligence Platforms

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ABSTRACT

Business Intelligence (BI) platforms are applications to analyze critical business data so as to gain new insights about business and markets. The new insights can be used for improving products and services, achieving better operational efficiency, improve competitiveness, and fostering customer relationships. However, we notice that BI is not yet a reality in Small and Medium Enterprises (SMEs) and it's even unknown in many of these organizations. The implementation of BI in organizations depends on their specificities. It is essential that the information available on platforms and their functionality comes clearly to organizations. Only a correct choice, in response to business needs, will allow powerful data that results in gains and business success. For this purpose, we have developed a comparative analysis of the existing capabilities in the various BI tools, which is intended to assist the selection of the BI platform best suited to each organization and/or business area. In this paper, we study and compare seven of the most used open source BI platforms: Actuate, JasperSoft, OpenI, Palo, Pentaho, SpagoBI and Vanilla.

Categories and Subject Descriptors

H.4.2 [Information Systems Applications]: Types of Systems – *Decision support*; J.1 [Computer Applications]: Administrative Data Processing; K.6.3 [Management of Computing and Information Systems]: Software Management – *Software Process*.

General Terms

Algorithms, Management, Measurement, Performance, Verification.

Keywords

Business Intelligence, BI Platforms, Open Source BI tools.

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ISDOC 2014, May 16 - 17 2014, Lisbon, Portugal.

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<http://dx.doi.org/10.1145/2618168.2618182>

1. INTRODUCTION

The concept of Business Intelligence (BI) refers to the process of collecting, organizing, analyzing, sharing and controlling data that provides support for organizational management activities. BI can also be described as the mechanism that "*provides actionable information delivered at the right time, at the right location, and in the right form to assist decision makers*"[1].

The genesis of the BI concept is attributed to Howard Dresner, when he was an analyst of the Gartner Institute. Initially, the concept meaning was associated with organizational management more than the technological area and referred to the set of models and methods that promoted decision making, using support systems sustained on data.

All the organizational context requires rigorous planning, regularity/uniformity of procedure and optimization of the overall existing resources. These assumptions require a constant and regular access to updated and relevant data, in order to decide which ones enable the sustainability and growth of organizations.

ACM 978-1-4503-2713-8/14/05\$15.00BI platforms enable companies to measure and improve the metrics that matter most to their businesses, such as sales, profits, costs, quality defects, safety incidents, customer satisfaction, on-time delivery and so on.

Consequently, the implementation of BI platforms for decision support in enterprises emerges as a pressing need, resulting from competitive dynamics practiced by large business organizations. This requires firms to make efforts in order to put in markets products and services at competitive prices.

Business Intelligence points to a set of management methodologies, implemented through software tools, whose function is to provide profitability and leadership in decision-making and administration of organizations. Thus, at the highest level of management, decision-makers have, in a given place and time, analytics tools which provide, as a group, important information and data.

At the present, the whole of organizations, particularly larger ones, enjoy of Information Technology (IT) systems that can collect, process, transmit and disseminate data. IT systems are designed to record large amounts of data generated by the business process.

As a result of the BI tools use, all these data acquire structure and organization, providing essential and fundamental information to decision-making. This is supported on the evidence generated by these tools, such as tables, graphs, dashboards, KPIs (Key

Performance Indicators), OLAPs (On-line Analytical Processing) and Data mining, among others. By interpretive reading of these elements, decision makers can perspective and deduce trends, patterns or deviations in the respective business area. This will promote the strengthening of the strategic planning of organizations, as well as proper management of contractual relationships with different partners, resulting in effective business growth. This growth cannot ignore the dynamics of social networks as a source of information about behaviors and associations in the markets and their agents. The conversion of such information into organizational knowledge, through the integration of collaborative technologies in BI turns out therefore, in our view, essential to the sustainability of companies.

This article aims to make a comparative analysis of existing functionalities in BI Open Source platforms and to highlight the fact that the universe of Small and Medium Enterprises (SMEs) reveals a lack of knowledge about these tools and their potential to business development. We believe it is essential to promote adequate disclosure and information about BI, with special focus on the associated benefits.

Many SMEs do not develop precise strategies in the IT area. They have little professionalized and a reduced profitability incipient reality. There are several reasons for this: lack of information, financial unavailability and/or little innovative management processes. It is in this context that there is a lack of potential of BI and competitive advantages for the positioning of companies in the market.

The remainder of this paper is structured as follows. Section 2 discusses the evaluation criteria used in the comparison and analysis of existing BI functionalities. Section 3 presents and characterizes the Open Source BI platforms to be analyzed. Section 4 presents a comparative analysis of existing functionalities in each BI Platform. Finally, in section 5 we present our conclusions and point out future work

2. EVALUATION CRITERIA

As starting point of our work we adopt the fifteen evaluation criteria of BI platforms used in Gartner [2]. These criteria are grouped into three broad and fundamental categories: Integration, Information delivery and Analysis.

The **Integration** category includes the BI infrastructure, metadata management, development tools and collaboration criteria.

The category **Information Delivery** contains reporting, dashboards, ad hoc query, Microsoft Office integration, search-based BI and mobile BI.

The **Analysis** category includes online analytical processing (OLAP), interactive visualization, predictive modeling and data mining, scorecards (KPI - key performance indicators), and prescriptive modeling, simulation and optimization.

In the Gartner's perspective the above criteria are functionalities that, once attached to BI platforms, enable organizations with clear systems of classification and measurement in order to give support to decision making. Consequently, organizations may enhance their performance, efficiency and profitability levels, developing a clear vision of the different business dimensions: customers, employees and products.

We add two new criteria to Gartner proposal: **Cloud** and **Collaborative Technologies**. The concept of Cloud (cloud computing) refers to the use of the processing power, storage and memory existing in the Internet. Thus, applications, data and services are available on the Internet and no longer confined to computers and servers of organizations. Despite the various associated advantages noticed (reduced equipment costs and operations, etc.) some questions remain as to safety guarantees.

The Collaborative Technologies are assuming a significant and growing importance for decision makers in organizations, because they enable the integration of data from social networks. These represent an invaluable and inexhaustible source of information to organizations on behavior, preferences, trends, expectations and needs, from the most different points of the planet. These data, once processed, can be decisive in strategic planning and business management.

Analyzing the results of a survey of 2300 CIOs [3], on its technological priorities, it was concluded that the Collaborative Technologies will integrate in the very near future and irreversibly, the BI platforms. According to the survey, the Collaborative Technologies rose from 11th place in 2010 to 4th place in 2012, shortly after Business Intelligence, Mobile Technologies and Cloud (SaaS, IaaS, Paas), which emphasize that these three are already part of the BI platforms.

This achievement of Collaborative Technologies in the priorities of the decision makers demonstrates how unavoidable it seems that they come to permanently integrate BI platforms. It should be noted, however, that they're already available in MicroStrategy [4] platform.

It's a fact that the BI Platforms are constantly and rapidly evolving, integrating new criteria corresponding to fluctuations and market trends. It's illustrative of this, as can be seen in afore mentioned CIOs survey, the ability to access BI on Mobile Devices (Android, iPhone, Windows Mobile). This functionality that in 2011 was not part of the criteria of the majority of BI platforms is, after two years, available on all platforms consulted.

Because, we consider the presence of Collaborative Technologies in BI platforms will be a requirement for organizations, in future work we will put the focus on them.

Back to the previously stated criteria Gartner, we selected nine of the fifteen existing: Reporting, Dashboards, Ad-hoc queries, Microsoft Office integration, Mobile BI, On-Line Analytical Processing (OLAP), Interactive Visualization, Predictive Modeling/Data Mining and Scorecards (KPI). The selection of these criteria was based on the fact that it is possible to measure the respective functionalities. The remaining criteria show some commercial constraints of access and verification. We will therefore succinctly expose, what functionalities each chosen criteria entails.

Reporting criterion refers to the ability to generate formatted and interactive reports, with or without parameters, with highly scalable distribution and possibility of scheduling capabilities.

Dashboards consist of the publication of reports on the computer, web or on mobile devices with intuitive, interactive dashboards that allow the evaluation and comparison of relevant information to achieve one or more business objectives. They allow an overview and an easier tracking of business. The use of these

panels is increasingly common to promote strategic, financial and operational data in real time.

Ad-hoc queries allow users to ask their own questions of the data, without relying on IT to create a report. In particular, the tools must have a robust semantic layer to enable users to navigate available data sources.

Microsoft Office integration criterion is based on the fact that Microsoft Office, especially Excel, is often used in the making of reports and analysis. In this context it is essential to ensure that this tool enables integration with Microsoft Office.

Mobile BI enables organizations to develop and deliver content to mobile devices in a publishing and/or interactive mode, and takes advantage of mobile devices' native capabilities, such as touchscreen, camera, location awareness and natural-language query. Regardless of the location of customers, employees or suppliers, continuously updated information is available. On the other hand, and from a commercial point of view, the fact that organizations can be aware of the customers location, via mobile devices, can be determinant in the design of management strategies in order to a major competitiveness.

Also relevant for enhancing competitive strategies, **On-Line Analytical Processing (OLAP)** enables users to analyze data with fast query and calculation performance, enabling a style of analysis known as "slicing and dicing." This capability could span a variety of data architectures (such as relational, multidimensional or hybrid) and storage architectures (such as disk-based or in-memory). The different dimensions of the business (sales figures, budgets, quantities sold, etc.) data at various levels, can be obtained with more or less detail depending on the needs of the user.

Interactive Visualization criterion enables the exploration of data via the manipulation of chart images, with the color, brightness, size, shape and motion of visual objects representing aspects of the dataset being analyzed. This includes an array of visualization options that go beyond those of pie, bar and line charts, including heat and tree maps, geographic maps, scatter plots and other special-purpose visuals. These tools enable users to analyze the data by interacting directly with a visual representation of it.

Predictive Modeling/Data Mining enables organizations to classify categories of variables, to estimate continuous variables and identify patterns that establish predictive models of events and scenarios using mathematical algorithms.

Scorecards (KPIs) consist of measurable performance indicator elements. Depending on the organization's strategic objectives, it becomes essential to select and prioritize key performance indicators (KPIs). These arise as measures from which it is possible to analyze the evolution of business, measure levels of efficiency and satisfaction, in order to enhance the business objectives.

These are the eleven capabilities that enable organizations to support decision making and improve performance. These criteria will be tested on the seven open source BI platforms described in the next section.

3. STUDY OF OPEN SOURCE BI PLATFORMS

Many authors have refereed the advantages of using open source software [5, 6, 7]. The first perceived advantage is the fact that open source software is made available gratis or at a low cost. However, what really distinguishes open source software from software available without fee is the combination of four essential freedoms [8]:

0. The freedom to run the program, for any purpose.

1. The freedom to study how the program works, and change it so it does our computing as we wish.

2. The freedom to redistribute copies so you can help our neighbor.

3. The freedom to distribute copies of our modified versions to others. By doing this we can give the whole community a chance to benefit from our changes.

Open source tools are distributed on several licenses. Each license beyond confirming the "openness" of the software tools, also defines the rules by which the source code may be amended, modified and then distributed, among many others rules. The open source licenses allow unrestricted and costless access to software previously developed [9, 10]. The open source tools allow a dynamic in community and can be adapted, reconfigured, improved and returned to unrestricted use.

The use of commercial platforms requires the payment of expensive licenses to the holders. This fact immediately imposes a limitation and restriction on access, considering the high costs involved. Only organizations with major financial capacity can acquire them. Because of that, in this paper our focus is only in open source BI platforms.

We studied the following open source BI platforms analyzed are: Actuate, JasperSoft, OpenI, PALO, Pentaho, SpagoBI and Vanilla.

3.1 Actuate

Actuate [11] provides software to developers and manufacturers who build scalable, secure solutions that save time and improve brand experience by delivering personalized analytics and insights of their customers, partners and employees. Actuate founded and supports the project Eclipse BIRT (Business Intelligence and Reporting Tools) – the open source IDE – and develops specific software for public institutions in partnership with the Eclipse Foundation¹.

This platform is available in version Open Source BIRT and Actuate BIRT Commercial.

Figure 1 illustrates the set of resources for the design and implementation of projects that the Eclipse BIRT open source contemplates and offers. It includes ad hoc query, reporting, on-line analytical processing (OLAP) and dashboards, and also allows Enterprise Information Systems (EIS), Scorecards (KPI), Predictive Modeling, Data Mining and Extract Transform Load (ETL) tools.

¹ The Eclipse Foundation is a nonprofit foundation that supports, develops and promotes open source software's.

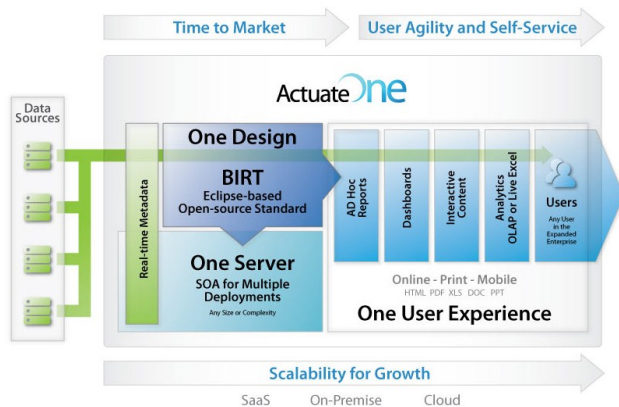


Figure 1 –Actuate One Platform

(Source: <http://www.actuate.com/download/whitepapers/BIRT-Analytics-Predictions-on-Big-Data.pdf>)

BIRT Analytics lets thoroughly explore huge amounts of data—petabytes of information from personally selected sources—and analyze the findings without IT help.

3.2 JasperSoft

The Jaspersoft [12] BI platform delivers reporting, dashboards, analysis, and data integration services for both stand-alone and embedded BI requirements, providing OLAP and in-memory data exploration.

The Jaspersoft platform is a SaaS (Software-as-a-Service), allowing fast transformation of data, reports and static graphics into interactive elements, with potential for immediate and automatic sharing by the entire company structure.

The final user can easily produce interactive reports, dashboard, analysis and exploitation of data, without the intervention of IT professionals. It also allows integration with iOS and Android mobile devices.

This platform is available under two types of licenses: GPL (Jaspersoft BI Community), which is free and Commercial (Jaspersoft BI Express, for AWS, Professional and Enterprise), with acquisition costs.

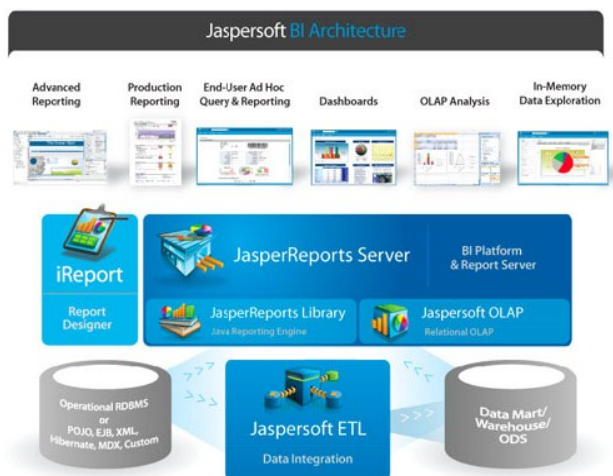


Figure 2 – Architecture Jaspersoft BI Platform

(Source: <http://www.dewcis.com/products/products-by-category/business-process-management/jaspersoft/>)

In Figure 2 is shown the architecture of Jaspersoft BI platform. At the center of figure are illustrated the main components of the platform: Jasper Reports Library - Java Reporting Library for developers, iReport Designer - Graphical Report Designer for developers and power-users, Jasper Reports Server - Interactive Report Server for business users and Jaspersoft OLAP - Interactive Data Analysis/OLAP for business users.

JasperReports Server provides reporting and analytics that can be embedded into a web or mobile application as well as operate as a central information hub for the enterprise by delivering mission critical information on a real-time or scheduled basis to the browser, mobile device, printer, or email inbox in a variety of file formats.

3.3 OpenI

OpenI [13] platform is the result of the release of a previous BI application under an Open-Source license in 2005. Unlike other platforms all the development is handled through SourceForge. The development model followed by the OpenI project is also the Free/Libre Open Source Software - FLOSS model, so there is only one edition of the software.

Figure 3 represents the architecture of OpenI platform. The range of BI functionalities are also illustrated: charting, creating Pivot tables, reports, OLAP and data mining processes.

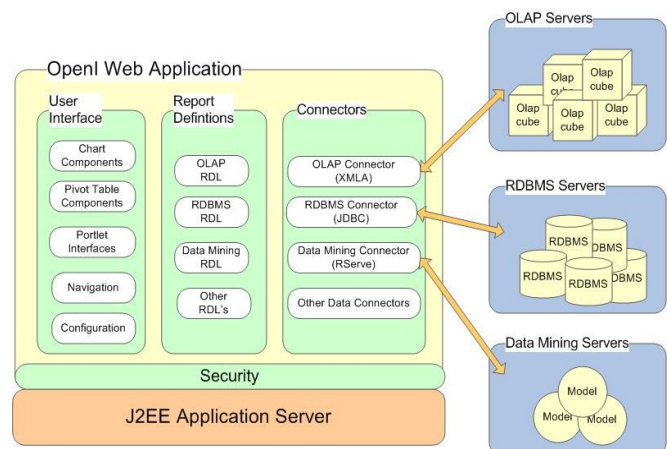


Figure 3 – Architecture OpenI

(Source: http://openi.sourceforge.net/openi_product.html)

Programmed in Java and using JSP technology, the components used by OpenI include the ubiquitous Mondrian OLAP server and the JasperReports reporting engine. OpenI doesn't include an ETL tool, or a Data Analysis component. The development team of this platform was more focused in the "Software as a Service, SaaS" concept than in the more usual Business Intelligence full-fledged application.

OpenI is a web-based OLAP reporting application. OpenI is an out-of-box solution for building and publishing reports from XMLA-compliant OLAP data sources such as Mondrian Microsoft Analysis Services or Mondrian.

With version 3.0 OpenI is deployed as a plug-in on top of Pentaho or Jaspersoft. However, in the site [9] is the information that platform was discontinued and they encourage users to migrate to using OpenI as a plug-in for Pentaho or Jaspersoft.

3.4 PALO

Palo [14] is a tool developed by the Jedox AG company, founded in 2002 and has a GNU license. Palo Suite combines all core applications — OLAP Server, Palo Web, Palo ETL Server and Palo for Excel — into one customizable Business Intelligence platform. The platform is completely based on Open Source products, which is available entirely free of any license fees.

The Palo was designed to support Microsoft Office Excel and Open Office Calc. The Palo for Excel includes a server of the OLAP database and the connection to Microsoft Excel and Open Office Calc is via an add-in that lets add a separator of BI functionality in the application (Excel or Calc) environment. This tab displays a range of functionalities that allows to work the registered data in worksheets, being the charts and reports generated by using the existing tools in the support application (Excel or Calc). In Palo platform, the inclusion of data ETL and OLAP queries are the functionalities that acquire higher expression.

Figure 4 gives an overview of Palo architecture and shows the interaction of multiple data sources with the ETL tool, the Palo OLAP Server, and subsequently with the Front End of the platform.

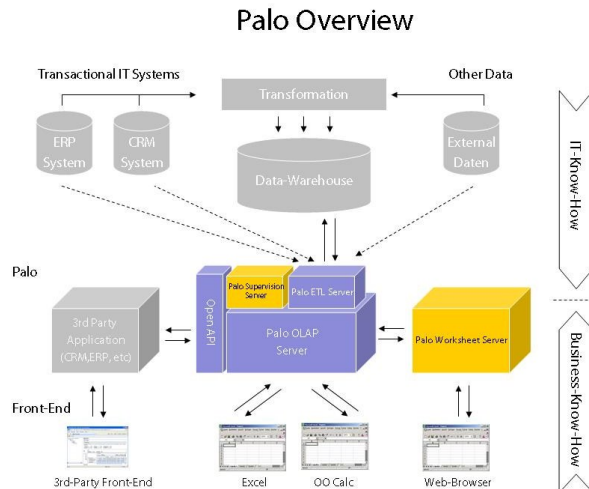


Figure 4 – Architecture Jedox Palo BI Suite

(Source: <http://www.legrandbi.com/2009/05/jedox-veut-faire-de-palo-la-solution-open-source-de-reference-pour-le-cpm/>)

Jedox OLAP Server also supports real-time modelling. This means users can modify hierarchies and even create new cubes on the fly in the familiar Excel environment. The database returns new aggregated values immediately. Real time modelling is a powerful simulation feature, that lets planners play out what-if scenarios involving changes in their corporate structures.

3.5 Pentaho

The Pentaho [15] platform follows the commercial Open Source model of development. Its implementation is controlled by Pentaho Corporation and is distributed in two distinct editions: The Community Edition, and the Enterprise Edition.

The Community Edition is the open source version while the Enterprise Edition is a commercial product that includes not only all the features of the Community Edition, but also some more

advanced tools to help the development of Business Intelligence Models and Reports, and a contract support.

Pentaho BI Platform is developed in Java language, which includes the following functionalities: ETL, OLAP, Data Mining and Predictive Modeling, reports and dashboards.

As can be seen in Figure 5, the platform consists of the following components: Pentaho Data Integration (through the open-source for ETL Kettle tool), Pentaho Analysis Services (also through an open source tool - Mondrian OLAP server), Pentaho Reporting, Pentaho Data Mining (using the Weka software), Dashboards and Pentaho for Apache Hadoop.

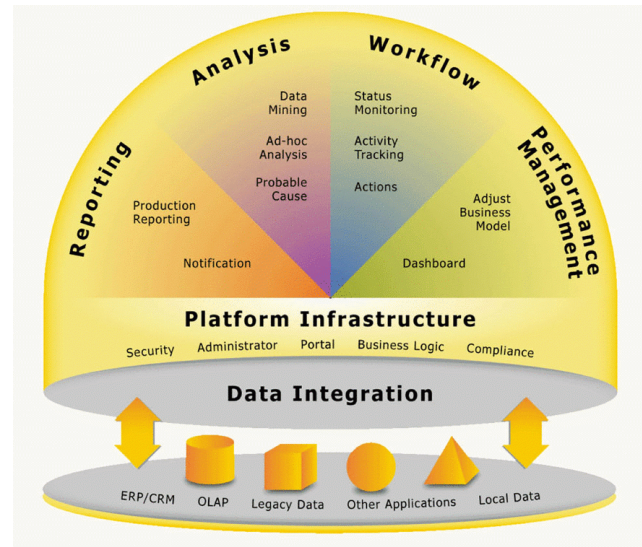


Figure 5 – Architecture of the Pentaho Open BI Suite

(Source: <http://www.bizcubed.com.au/pentaho-pillars>)

The Pentaho BI platform provides the architecture and infrastructure required to build solutions to business intelligence problems. This framework provides core services including authentication, logging, auditing, web services, and rules engines.

3.6 SpagoBI

SpagoBI [16] is an application developed by SpagoWorld, an organization founded in 2006 with the support of an open source community.

The SpagoBI tool has a GNU LGPL (Lesser General Public License), which allows users to use all the functionalities. SpagoBI suite is 100% open source, adopting the pure open source model and is released as only one stable version. It is an alternative to the "dual-licensing" or "open core" model, which also includes a parallel proprietary software release as an enterprise or professional version. The SpagoBI is developed in Java and available only on a completely free version.

The SpagoBI tool contains a variety of functionalities such as the creation and export of reports, creating charts, ETL, OLAP, Data Mining, KPIs, application server, login service, Dashboard, task schedule and Web server.

In Figure 6 we can see the modular structure of SpagoBI. All modules are connected with the core, which ensures the balance of the platform, as well as its ability to develop.

SpagoBI is a complete BI platform, robust, secure, and scalable. SpagoBI has unique solutions, not only usual reporting and charting tools but also with the following functionalities: Geo, Query by Example (QbE), KPIs, Interactive Dashboards, and Real-time BI.

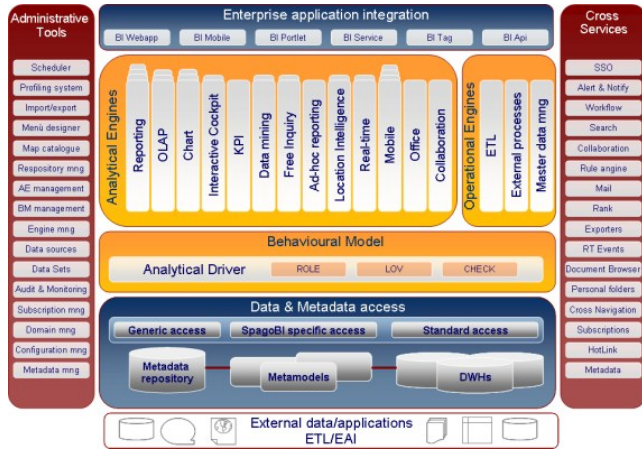


Figure 6 – SpagoBI Server

(Source: <http://www.spagoworld.org/xwiki/bin/view/SpagoBI/SpagoBIserver>)

3.7 Vanilla

The French enterprise BPM-Conseil founded in 2004, are the French leader of Open Source BI and responsible for developing of Vanilla platform [17]. Vanilla is 100% Open Source and has a MPL Licence (Mozilla Public License). It is written in Java and the Web Portal in GWT (Google Web Toolkit) technology.

Vanilla platform comes with all necessary tools to manage data from applicative software to the end user visualization. The last version is Vanilla 4.4.2 that is composed of several components: Web Portal, Free Analysis Web, FreeWebReport, FreeDashboard, Architect, BiGateway, BiWorkflow, BiProcessManager, FreeMetadata, Vanilla plugins for Birt, FreeDashboard, FreeAnalysis Schema Designer, Enterprise Services and Norparena.

Vanilla now offers support for the latest database version from leading editor such as MySql (version 5.6), MariaDb (10), Oracle (version 12), PostgreSQL (version 9.3) and Microsoft SqlServer2008 / SqlServer2012.



Figure 7 - Vanilla specificity

(Source: http://www.bpm-conseil.com/content/vanilla-presentation#services_technologies_title)

In Figure 7 are shown the different features and characteristics of the platform such as integrations with ETL, Key Performance Indicator Database, SaaS mode, allowing hosting of different projects, among others.

4. COMPARATIVE ANALYSIS

Based on the technical specifications of BI platforms available in the market and analyzed according to the criteria defined in section 2, we drawn up a comparative table on the availability of respective capabilities, taking into account their characteristics.

The table is intended as an aid to selecting the most suitable open source BI platform to an organization, depending on the needs that feel of its business area.

It wasn't our purpose to evaluate the usability of the tools functionalities'. That would require an individualized and case to case analysis, with the selection of only a few BI tools, according to particular organizational context. We're not therefore making any judgment on the tools performance, but revealing what each is able to do, enabling an option that run into the particularities of organizations. Thus, it can be implemented, taking in consideration lowering the costs, the BI platform most appropriate to the business.

Some open source platforms have commercial versions with a large number of functionalities, which determines the respective community versions exhibit a reduced amount of functionalities. For example, Jaspersoft platform only validates 7 of the 11 criteria in the community version and in the commercial version - Enterprise - validates almost all of the criteria.

Table 1 – Open Source BI Platforms Functionalities

Functionalities / Criteria	Business Intelligence Platforms						
	Actuate	JasperSoft	OpenI	PALO	Pentaho	SpagoBI	Vanilla
Reporting	✓	✓	✓	✓	✓	✓	✓
Dashboards	✓	✓	✓	✓	✓	✓	✓
Ad-hoc Queries				✓	✓	✓	✓
Microsoft Office Integration	✓	✓		✓	✓	✓	✓
Mobile BI		✓			✓	✓	✓
OLAP	✓	✓	✓	✓	✓	✓	✓
Interactive Visualization		✓	✓	✓	✓	✓	✓
Predictive Modeling/Data Mining			✓		✓	✓	✓
KPIs					✓	✓	✓
Collaborative Technologies						✓	
Cloud	✓	✓		✓	✓	✓	✓

Analyzing only the open source versions (community), it appears that platforms from Pentaho and Vanilla present a validation of 91% supporting 10 of the 11 criteria evaluated. The SpagoBI validates 100%. They are, therefore, very competitive platforms when compared with commercial, maybe even an effective alternative to these. However, from these three platforms,

SpagoBI and Vanilla is a “pure” open source with just a community version. It is important to remember that this study does not focus on the comparative analysis of the effectiveness of the platform functionalities. So at present, we cannot show results on their performance, based on criteria such as speed, usability, reliability, robustness, relationship investment/return, etc.

All the platforms include the criteria Reporting, Dashboards and OLAP. The presence of Reports and Dashboards is justified by the fact that they are the most used management tools in organizations. Similarly, OLAP allows end users to analyze data with fast searches and multidimensional views, which is a benefit that determines their integration in BI platforms.

Contrary to these functionalities, the collaborative technologies are only present in one platform, the SpagoBI. This is due to the fact that it is a relatively recent requirement of decision-makers and executives organizations. We reiterate our perspective that all BI platforms will come shortly to integrate this functionality in response to markets.

5. CONCLUSIONS AND FUTURE WORK

Business Intelligence has two capabilities essential to organizations. On one hand, it enables cost reduction, and secondly, it increases revenue. This happens because BI tools can be a vital support to management strategies and decisions. The BI provides data analysis that can help to create, improve or redefine products, services and/or processes of marketing and trading. This way, any organization or company, regardless of size, business area or positioning in domestic and international markets, can benefit from the BI implementation.

The selection process of BI platform best suited to an organization will depend on many factors like the existing financial, human and material resources, but essentially on the vision that the CEO and/or entrepreneur has to the organization.

The selection and implementation of BI will have to start from a correct diagnosis of the organizational context; a survey of the conditions, resources, weaknesses and potential should be made. Next, there's the planning and scheduling of all phases to develop, the tools implementation, their operationalization and the monitoring of the results to, if necessary, make adjustments.

It is essential that companies, especially Small and Medium Enterprises, understand the investment in BI as a synonymous of new business opportunities. Thus, is of great importance for these companies to access the open source BI tools.

Alike the mobile devices, which rapid massification determined their integration in almost BI platforms, answering to the markets, we assume that, in a short time, the platforms producers will decide upon the integration of Collaborative Technologies in BI tools.

From our analysis, we conclude that open source Business Intelligence platforms are growing both in features, quality and visual appeal. Of the seven tools analyzed, we cannot say which one is the true winner but Pentaho, SpagoBI and Vanilla tools are

those with the greatest potential for use in Small and Medium Enterprises.

As future work, we intend to extend the comparative analysis of existing functionalities in open source platforms to BI commercial counterparts.

6. ACKNOWLEDGMENTS

Our thanks to ISEP, ISEC, GECAD and CISUC for their support.

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